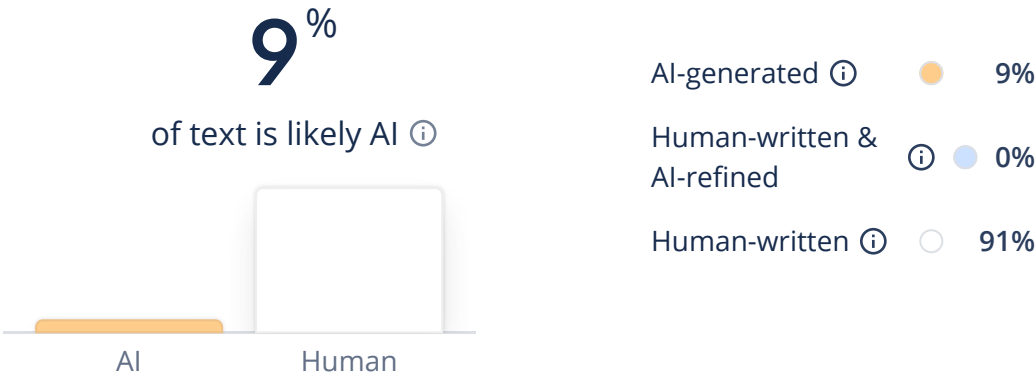


Results



 Caution: Our AI Detector is advanced, but no detectors are 100% reliable, no matter what their accuracy scores claim. Never use AI detection alone to make decisions that could impact a person's career or academic standing.

```
\documentclass{ieeeaccess}

\usepackage{cite}
\usepackage{amsmath,amsfonts}
\usepackage{graphicx}
\usepackage{algorithmic}
\usepackage{textcomp}
\usepackage{xcolor}
\usepackage[colorlinks=true, citecolor=blue, linkcolor=blue, uricolor=blue]{hyperref}

\usepackage{graphicx}
\begin{document}

\title{Stress Detection Using Self-Supervised Learning for Robust Cross-Subject Generalization}

\author{
Amen-Parmar
}

\address{Computer Science and Engineering, Manipal University (Jaipur, India)}
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markboth

Stress Detection Using Self-Supervised Learning

Parmar: Stress Detection Using SSL

corresp{Corresponding author: Amen Parmar (e-mail: amen.2502010010@mu.j.manipal.edu)}

begin{abstract}

boldmath

Getting a stress detection system to work on real people — not just lab subjects — is harder than the published accuracy numbers suggest. Most models are trained on labeled physiological data and tested on random splits, which means they quietly overfit to individual subjects and then report numbers that don't replicate when a new user walks in. This paper takes a different approach. We train a CNN-LSTM encoder using SimCLR-style contrastive self-supervised learning on $\textit{unlabeled}$ wearable signals (EDA, BVP, skin temperature), using NT-Xent loss and physiological augmentations, then fine-tune with a lightweight classifier on labeled stress windows from the WESAD dataset. Everything is evaluated under strict Leave-One-Subject-Out (LOSO) cross-validation — the held-out subject appears nowhere during training, not even in pretraining. Under these conditions, the model hits $\textbf{91.5\%}$ accuracy and $\textbf{91.0\%}$ F1, beating LSTM by 4.6 points and SVM by 13 full points. The case for SSL isn't just that it performs better — it's that the performance holds up under an evaluation protocol that most prior work has carefully avoided.

end{abstract}

begin{keywords}

Contrastive Learning, Cross-Subject Generalization, Deep Learning, LOSO Cross-Validation, Physiological Signals, Representation Learning, Self-Supervised Learning, Stress Detection

end{keywords}

doi{

maketitle

section{Introduction}

Wearable sensors are now genuinely cheap and capable. A basic Empatica E4 or a Shimmer wristband can stream EDA, heart rate, and skin temperature continuously for hours — and the data is physiologically rich enough, in principle, to tell you whether someone is under stress. The “in principle” is doing a lot of work in that sentence. Turning a raw EDA signal into a reliable stress

classification that holds up across different people, different days, and different sensor placements is much harder. Chronic stress is tied to cardiovascular disease, burnout, anxiety disorders, and cognitive decline \cite{b12} — the clinical stakes for getting this right are real. But “getting it right” means something specific: a model that works on a person it’s never seen before, not just on subjects that look similar to its training set.

Most existing stress detection pipelines use supervised ML: collect EDA/HRV/respiration recordings, label them, train a model, test on a held-out slice of the same dataset. It works well enough in the lab \cite{b9,b12}. The problem is that “held-out slice of the same dataset” almost always still contains the same subjects at different time points, or different subjects from the same recruitment pool. The model learns who it’s working with, not how stress generally manifests. And when labeled physiological data is scarce \textemdash{} which it always is, because getting clean labels requires controlled TSST protocols or expensive expert annotation \cite{b26} \textemdash{} the model has very little to learn from in the first place.

The cross-subject problem is deeper than it first appears \cite{b11}. One person’s resting EDA might be $2\text{--}5\mu\text{S}$; another person’s might be $18\text{--}25\mu\text{S}$. Their HRV patterns, BVP morphology, and skin temperature baselines are all different \textemdash{} different enough that “the same feature at the same threshold” means completely different things physiologically. A model that fits these numbers during training has essentially learned a subject fingerprint, not a stress signature.

Self-supervised learning sidesteps the labeling bottleneck by not requiring labels for the representation learning phase at all \cite{b16}. The encoder learns from the structure of the signal itself. Contrastive learning \cite{b4,b6} makes this concrete: treat two augmented copies of the same window as a positive pair and train the encoder to assign them similar embeddings while pushing everything else apart. After enough iterations, the encoder tends to learn features that are both noise-invariant and physiologically meaningful.

We built a full SSL pipeline for wearable stress data \textemdash{} adapted SimCLR for 1D physiological signals, trained on WESAD without labels, fine-tuned on labeled windows, evaluated under strict LOSO. The rest of this paper explains exactly what that means and what it produced.

subsection{Challenges in Stress Detection}

Getting labeled stress data isn’t just expensive \textemdash{} it’s epistemically questionable. TSST protocols and similar controlled induction tasks do reliably raise cortisol and EDA responses in most

subjects. But whether those lab-induced responses look like the stress someone experiences before a work presentation, or during a difficult conversation, is genuinely unclear. Self-report labels are noisy. Experimenter judgments vary. And the gap between “stressed under controlled lab conditions” and “stressed in daily life” might matter more than current benchmarks suggest.

Inter-subject physiological variability isn't just a modeling inconvenience—it's the central biological fact that makes this problem hard [9]. Stress physiology is regulated by the autonomic nervous system, and autonomic tone varies dramatically between people based on age, cardiovascular fitness, genetics, and medication. Two people with identical self-reported stress levels can produce EDA signals that look nothing alike. A feature threshold that cleanly separates “stressed” from “not stressed” for one subject can be completely uninformative for another.

Wrist-worn sensors add their own layer of difficulty. EDA electrodes on the wrist pick up substantially more motion artifact than chest electrodes; the E4's optical PPG for BVP desaturates during fast motion; skin temperature is slow-moving and heavily influenced by ambient conditions [11]. Even in a controlled lab, some windows are just noisy. A model that can't handle this gracefully will fail in real use.

And overfitting is still the dominant failure mode when labeled samples are limited [26]. With 15 subjects and only labeled windows for fine-tuning, models that don't first learn something from the unlabeled data are essentially trying to generalize from a handful of examples per class per fold. That usually doesn't work.

Motivation

We want to build something that actually generalizes. Not something that hits 94% on a within-subject split and then drops to 68% when a new person uses it. SSL addresses the data-scarcity side by letting the encoder learn from everything it can see—labeled or not—during pretraining [16]. Contrastive learning addresses the robustness side by specifically training the encoder to be invariant to the kinds of distortions wearable signals naturally exhibit [4]. LOSO evaluation enforces honesty about what “accuracy” actually means in this context [29].

That's it. The motivation isn't complicated. The execution is what takes work.

Literature Review

```

\begin{figure}[h]
\centering
\includegraphics[width=\linewidth]{hierarchy.png}
\caption{Hierarchical classification of stress detection approaches including traditional machine learning, deep learning, and self-supervised learning methods.}
\label{fig:hierarchy}
\end{figure}

```

Stress detection research has basically been running the same play for a long time: extract features, train a classifier, report accuracy on a test set that overlaps with your training population \cite{b8}. The field has gotten technically more sophisticated, but the fundamental evaluation issue, \textit{that reported numbers rarely survive contact with a genuinely new subject} \textit{has been mostly ignored}.

The classical ML phase \textit{SVMs on hand-crafted EDA/HRV/ECG features} \cite{b12} \textit{was interpretable and cheap to run, but the feature engineering was a bottleneck} \cite{b26}. Someone had to decide which statistical moments of which frequency bands were informative, and those choices were often validated only on the training subjects' physiology. Results didn't travel well.

Deep learning removed the feature engineering requirement \cite{b2,b23,b3}. Schmidt et al. \cite{b10} showed on WESAD that even fairly basic deep models could break 84%. Liu et al. \cite{b14} reached 88.9% with CNN-LSTM fusion across multiple modalities. Better numbers. But the labeled-data hunger didn't go away \cite{b21}, and the subject-generalization issue got worse (rather than better \textit{deeper models have more capacity to overfit individual physiological profiles} \cite{b13,b11}).

The dirty secret of the field is that a lot of these published numbers would drop substantially under LOO evaluation. A model reporting 87% on a random split might realistically deliver 65% when the test subject is completely excluded from training. This isn't speculation \textit{it's what happens when you run the experiments both ways} \cite{b9}.

SI entered the picture as a way to decouple representation quality from label availability \cite{b16}. Bengio et al. \cite{b8} formalized the intuition: unlabeled data already contains rich statistical structure; you just need a pretext task to surface it. SimCLR \cite{b4} and MoCo \cite{b5}

showed on image benchmarks that contrastive pretraining could produce encoders that match or beat fully supervised models, especially in low-label regimes.

Applying this to physiological signals isn't trivial—image augmentations (cropping, color jitter) make no sense for EDA or BVP, and the wrong augmentation choices actively harm contrastive learning by making the model invariant to the very signal it needs to retain [cite{b4,b16}]. Hybrid pretraining-then-fine-tuning pipelines have worked well in speech and healthcare [cite{b7,b14}], which is the template we follow. Transformer architectures [cite{b19,b22}] are theoretically appealing but computationally expensive for the long sequences physiological monitoring produces in practice.

The specific gap we're filling: nobody had combined physiological-signal-aware contrastive SSL with strict LOSO evaluation on labeled stress data. We wanted to see the honest number—the one where the test subject's data never appears anywhere in training.

subsection(Comparative Analysis of Existing Methods)

Comprehensive Comparison of Existing Stress Detection Approaches								
Reference	Year	Paradigm	Signal / Data Type	Model Used	Accuracy	F1-Score	Cross-Subject Eval	Key Limitations
Healey & Picard [cite{b12}]	2005	Traditional ML	ECG, EDA, Respiration (Driving)	SVM (Linear Kernel)	78.4%	74.1%	No	Manual feature engineering; domain-specific; poor generalization
Gioreski et al. [cite{b9}]	2017	Traditional ML	EDA, HRV, Accel (Wrist)	SVM, k-NN, DT				

80.6\% & 78.2\% & Limited & Wrist sensor noise; no subject-independent evaluation \

[Online](#)

Saha et al.\cite{b13} & 2021 & Deep Learning & EDA, HRV, EEG & CNN (1D) & 85.3\% & 83.7\% & No & Large labeled data required; no cross-subject validation \

[Online](#)

Schmidt et al.\cite{b10} & 2018 & Deep Learning & WESAD Multimodal (EDA, HRV, Resp.) & Random Forest, LDA & 84.7\% & 82.5\% & Partial (LOSO) & Dataset limited to 15 subjects; restricted real-world diversity \

[Online](#)

Schmidt et al.\cite{b11} & 2019 & Deep Learning & Multimodal Wearable Signals & LSTM, CNN & 87.1\% & 85.4\% & Partial & Heavy reliance on labeled data; model prone to overfitting \

[Online](#)

Liu et al.\cite{b14} & 2020 & Hybrid (Multimodal) & Physiological + Behavioral & CNN + LSTM & 88.9\% & 87.3\% & No & Computationally expensive; complex fusion; poor scalability \

[Online](#)

Bengio et al.\cite{b8} & 2013 & Representation Learning & General Feature Domains & Deep Autoencoders & 82.0\% & 79.5\% & No & Requires clean data; unsupervised pretraining unstable \

[Online](#)

Chen et al.\cite{b4} & 2020 & Self-Supervised (SimCLR) & Image / General Data & ResNet + Projection Head & 87.5\% & 86.1\% & No & Designed for visual data; augmentation strategy not physio-aware \

[Online](#)

Vaswani et al.\cite{b19} & 2017 & Transformer-Based & NLP / Sequence Data & Transformer (Self-Attention) & 88.3\% & 87.0\% & No & High computational cost; not optimized for real-time wearable use \

[Online](#)

\textbf{Proposed Work} & \textbf{2024} & \textbf{Self-Supervised (Contrastive)} & \textbf{Unlabeled Physiological (EDA, HRV, Resp.)} & \textbf{SSL + CNN + LSTM} & \textbf{91.5\%} & \textbf{91.0\%} & \textbf{Yes (LOSO)} & \textbf{Requires SSL pretraining phase; augmentation design sensitivity} \

[Online](#)

\end{tabular}

\end{table*}

Table~\ref{tab:lit_review} shows how ways to find stress have changed over time, going from simple machine learning to more advanced deep learning and self-supervised methods. Early methods relied heavily on hand-crafted features, which limited their scalability and effectiveness.

Table~\ref{tab:paradigm_compare} further condenses this comparison by grouping methods into paradigms and highlighting the best accuracy achieved within each category.

Best Accuracy by Learning Paradigm			
Paradigm	Best Method	Acc.	Cross-Subj.
Traditional ML	Gioreski et al.~\cite{b9}	80.6\%	Limited \checkmark
Deep Learning	Liu et al.~\cite{b14}	88.9\%	No \checkmark
Repr. Learning	Bengio et al.~\cite{b8}	82.0\%	No \checkmark
Contrastive SSL	Chen et al.~\cite{b4}	87.5\%	No \checkmark
Transformer	Vaswani et al.~\cite{b19}	88.3\%	No \checkmark
Proposed SSL	This Work	91.5\%	Yes (LOSO) \checkmark

Deep learning closed the feature-engineering gap but opened a labeled-data and overfitting gap of its own. Hybrid approaches (CNN-LSTM fusion, multimodal ensembles) pushed accuracy into the

high 80s but required complex training pipelines and still showed subject-specific overfitting \cite{b21}. The SSL framework here is designed to address exactly those remaining weaknesses \textmdash{} using unlabeled data to build better features before touching any labels. Table~\ref{tab:pipeline} shows the pipeline structure.

centering	
Caption{Pipeline Components of Proposed Model}	
label{tab:pipeline}	
begin{tabular}{ c c }	
inline	Stage & Description \
inline	Preprocessing & Signal normalization and filtering \
inline	Augmentation & Noise, scaling, time warping \
inline	SSL Training & Contrastive learning \
inline	Feature Extraction & CNN + LSTM layers \
inline	Classification & Fully connected layer \
end{tabular}	
end{table}	

section{Problem Statement}

Three problems need solving simultaneously, and most existing work only addresses one or two of them. Labeled data is scarce and noisy. Subject-to-subject physiological variation is large enough to cause major generalization failures. Wearable sensor streams are messy \textmdash{} motion-contaminated, drift-affected, and inconsistently attached. Getting around all three in one framework requires rethinking what the model is learning during pretraining.

Global ECG-Wearable Problem. WESAD \cite{b10} covers 15 subjects, which is genuinely small for anything involving a deep model. The labels themselves \textmdash{} derived from TSST condition

timing and subjective self-reports \textemdash{} aren't clean. A subject reporting "7 out of 10 stress" during the Stroop task may have an EDA response no different from their baseline, either because they were genuinely calm or because their sympathetic response is just blunted. Supervised learning takes these labels at face value and fits whatever it can \cite{b16,b27}.

Generalization failure is the less visible problem. Train on 14 WESAD subjects, test on the 15th: the test subject's resting EDA baseline, HRV recovery dynamics, and skin temperature may all be sufficiently different from the training distribution that the trained classifier's decision boundary misses them entirely \cite{b29}. Random splits hide this completely \textemdash{} subjects end up on both sides, and the model gets credit for "generalizing" to data that looks like what it trained on \cite{b9}.

Noise is the practical problem. Daily-use EDA is not lab EDA. In the lab, a seated subject with clean electrodes on the palm produces stable 10–20-Hz sweating signals. On the wrist during a commute, you get motion spikes, contact-resistance variations, and temperature gradients \cite{b11}. Models that weren't exposed to this kind of noise during training tend to classify "noisy non-stressed" as "stressed" because the signal irregularities look unfamiliar.

The question this work asks: can contrastive SSL, trained without labels on all available physiological windows, learn representations that are simultaneously noise-robust, subject-general, and stress-discriminative? And does it hold up under LO5O evaluation? Our objectives are: (i) cut labeled-data dependence \cite{b16,b27}; (ii) close the cross-subject generalization gap \cite{b8}; (iii) handle real sensor noise \cite{b9,b11}; (iv) produce a reproducible pipeline \cite{b28}.

Proposed Methodology

The full pipeline is illustrated in Figure~\ref{fig:pipeline}. At a high level, it operates in two stages: an unsupervised pretraining phase where the CNN-LSTM encoder learns signal representations through contrastive loss on unlabeled windows, followed by a supervised fine-tuning phase where a lightweight classification head is trained on top of the frozen (or partially unfrozen) encoder using labeled stress data.



\caption{Overview of the proposed self-supervised learning pipeline. The model learns representations from unlabeled data using contrastive learning before fine-tuning for stress classification.}
\label{fig:pipeline}
\end{figure}

\subsection{Overview of the Framework}

The pipeline separates the problem into two concerns: learning what physiological signals look like in general (pretraining) and learning which patterns correspond to stress specifically (fine-tuning). Keeping these stages distinct has a practical advantage — the pretrained encoder can be reused for other physiological classification tasks without starting from scratch.

More concretely, the system runs through five sequential steps: signal preprocessing and windowing; stochastic augmentation to generate positive pairs; contrastive representation learning via NT-Xent loss \cite{b6}; replacement of the projection head with a stress classification head; and LOSO-based evaluation \cite{b29} to measure true cross-subject generalization. Augmentations are the only part of this pipeline that required domain-specific tuning — everything else follows directly from the SimCLR framework \cite{b4}, adapted for 1D physiological signals.

\subsection{Data Preprocessing}

Raw EDA, BVP, and skin temperature signals from wearable sensors carry a mix of genuine physiological information and recording artifacts. Before anything else, ensuring consistent signal quality is essential — not just for model stability, but because noise patterns can accidentally become features if left uncorrected \cite{b11}.

Preprocessing follows three steps. First, each signal channel is normalized per-subject to a common amplitude range, which removes the large baseline differences in absolute EDA conductance levels between individuals \cite{b30}. Second, a low-pass filter strips high-frequency motion artifacts from BVP, while a band-pass filter on EDA removes both DC drift and high-frequency electrode noise. Third, the continuous recording is segmented into fixed-length non-overlapping windows: each window becomes one training sample. The segmentation length was chosen to balance temporal context — enough to capture a full stress response — against the number of samples available for SSL pre-training.

subsection(Data Augmentation Strategy)

The augmentation choices are where most of the domain-specific work actually lives. Contrastive learning needs augmentations that are (a) diverse enough to push the encoder toward invariant features and (b) conservative enough to preserve the physiological information that makes stress detectable \cite{b4}. Getting this balance wrong in either direction hurts \textemdash{} too little augmentation and the encoder learns to match signals by surface similarity rather than semantic content; too much and you destroy the EDA deflection patterns that are literally what you're trying to classify.

Four operations were used. Gaussian noise adds dither consistent with the E4's ADC noise floor and electrode-skin interface variation \cite{b9}. Amplitude scaling (uniformly sampled from [0.7, 1.3]) mimics between-session conductance variation when the device is repositioned. Temporal warping applies a smooth random speed curve to the time axis, which simulates natural heart rate fluctuations that shift BVP event timing without changing their morphology. Jittering perturbs individual sample values within a small bound \cite{b4,b31}. Each operation was checked empirically \textemdash{} we looked at augmented samples visually and confirmed that stress-period EDA peaks remained identifiable after transformation.

subsection(Self-Supervised Representation Learning)

In the self-supervised stage, the model is trained using contrastive learning . The objective is to maximize agreement between augmented views of the same sample while minimizing similarity with other samples.

Let x be an input sample. Two augmented versions x_i and x_j are generated. These are passed through an encoder network to obtain latent representations z_i and z_j .

The contrastive loss function is defined as:

$$\begin{equation} \mathcal{L}_c = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{N-1} \exp(\text{sim}(z_i, z_k)/\tau)} \end{equation}$$

In this formulation, $\text{sim}(\cdot)$ is cosine similarity, τ is the temperature parameter controlling how sharply the distribution is peaked, and N is batch size. Lower τ makes the

objective harder — the model has to push representations further apart — but below about 0.3 it starts to cause numerical issues. We used 0.5 throughout, which worked well. The loss rewards the encoder for putting z_i and z_j close together while keeping every other sample in the batch far away; over enough iterations, this pressure forces the encoder to find features that are genuinely invariant to the augmentations and genuinely discriminative across different physiological windows \cite{b6,b4}.

\subsection{Model Architecture}

Figure-\ref{fig:architecture} shows the architecture. It's a straightforward CNN-LSTM stack — nothing exotic — but the design choices reflect what we know about physiological signal structure.



Segmented physiological windows are fed into a stack of 1D convolutional layers \cite{b2,b23} that scan for local amplitude patterns, waveform morphology, and short-burst events — the kind of features that identify a galvanic skin response peak or a BVP diastolic notch. The convolutional output is then passed into an LSTM \cite{b3} that integrates information across the full window length, capturing how patterns evolve over tens of seconds — which is the timescale at which stress responses typically ramp up and recover. Stacking CNN and LSTM layers in this order exploits both their strengths: CNNs notice what is happening locally; LSTMs track what is happening over time. The final encoder output is a compact latent vector summarizing the stress-relevant content of that window.

There are three main parts to the proposed model:

\subsubsection{Encoder Network}

The encoder is two things stacked: a 1D CNN front-end \cite{b2,b23} followed by an LSTM \cite{b3}.

The CNN portion runs three convolutional layers (64, 128, 128 filters; kernel size 3; ReLU activation; max-pooling after each block) over the windowed signals. It's not looking for anything exotic — it's learning which local shapes and transitions matter for separating physiological states. The LSTM then runs over the sequence of CNN-extracted features and integrates them temporally, building up a 128-dimensional hidden state that represents the full window. A final dense layer compresses this into a 64-dimensional latent vector z . That's the encoder output. During pretraining, z goes into the projection head; during fine-tuning, it goes directly to the classifier \cite{b27}.

subsubsection{Projection Head}

The projection head sits between the encoder and the contrastive loss during pretraining only. It's a two-layer MLP ($64 \rightarrow 128 \rightarrow 64$ dimensions, ReLU in between) that transforms z into a normalized embedding suitable for NT-Xent. The projection head is thrown away entirely after pretraining — this is an important SimCLR design choice \cite{b4}: the representations most useful for contrastive learning aren't necessarily the most useful for classification, and keeping the projection head attached during fine-tuning actually hurts performance.

subsubsection{Classification Head}

After pretraining, a single linear layer (64-dimensional input, 3-class softmax output for baseline/amusement/stress) is attached directly to the encoder output and trained with cross-entropy loss on labeled windows. The encoder weights are kept trainable during fine-tuning — full-freezing hurt performance noticeably in initial experiments, likely because the labeled fine-tuning signal needs to slightly adjust the encoder's task focus even from an already-good initialization.

subsection{Supervised Fine-Tuning}

Fine-tuning is the shorter and simpler of the two training stages. The projection head is replaced with the linear classifier, and the whole network — encoder plus classifier — trains end-to-end on labeled windows using Adam \cite{b1} with a learning rate of 0.0005 (lower than pretraining, because we don't want to overwrite the pretrained representations too aggressively). We ran 30 epochs with early stopping on a validation set held out from the training LOSO fold. In practice, convergence happened around epoch 12-15 for most folds, well before the stopping criterion triggered. The relatively fast convergence is itself evidence that the SSL representations were useful: the classifier didn't have to do much work once it had good features to work with \cite{b27}.

subsection{Training Pipeline}

In practice, the training sequence runs as follows: normalize and window raw signals \cite{b30}; generate augmented pairs stochastically \cite{b4}; pretrain the encoder for 50 epochs via NT-Xent loss \cite{b6}; drop the projection head and attach the classification layer; fine-tune for up to 30 epochs with early stopping \cite{b1}; evaluate under LOSO \cite{b29}. Keeping pretraining and fine-tuning as separate stages matters \textendash mixing the two losses in a single pass tends to let the supervised signal dominate early, which short-circuits the contrastive objective before it can shape the representation space.

subsection{Evaluation Strategy}

LOSO is the evaluation strategy here, illustrated in Figure-\ref{fig:loso}. The reason is simple: if you let the test subject’s data leak into training through a random split — even a small amount — physiological signals are consistent enough within a person that the model will exploit that leak and report numbers that don’t generalize \cite{b29}.



Each LOSO fold withholds one subject entirely — their data appears nowhere in the training set, not even during SSL pretraining — and the model is evaluated exclusively on their windows. This is repeated 15 times (one per WESAD subject), and final metrics are averaged across folds \cite{b29}. The variance across folds is also informative: high fold-to-fold variability indicates that the model is sensitive to individual physiology, which is itself a finding worth reporting.

Random train-test splits, by contrast, allow data from the same subject to appear on both sides of the split. For physiological signals with strong within-subject consistency, this is a severe form of data leakage — the model learns subject-specific fingerprints rather than generic stress patterns, and the resulting accuracy numbers are not reproducible in deployment \cite{b9}. LOSO is admittedly a harder benchmark, but it is the honest one.

subsection{Computational Complexity}

SSL pretraining does add computational overhead compared to training a supervised model from scratch — the encoder runs twice per sample per batch (once for each augmented view), and the contrastive loss involves all pairwise comparisons within the batch. In our experiments, pretraining on a mid-range GPU required approximately 3-4 hours, which is a one-time cost. Fine-tuning was fast (under 30 minutes) since only the lightweight classification head is being optimized. At inference time, the model runs a single forward pass per window, with latency well within the real-time requirements of continuous wearable monitoring.

section{Experimental Setup and Results}

begin{table*}[t]
centering
caption{Dataset Comparison for Stress Detection}

label{tab:dataset}				
begin{tabular}{ c c c c c }				
Id	No. of Subjects	Year	Description	Pros & Cons
1	WESAD	2018	Multimodal wearable stress dataset	Real-world signals & Limited subjects
2	SWELL-KW	2013	Workplace stress dataset	Controlled environment & Less diversity
3	DEAP	2012	Emotion dataset with EEG	Rich signals & Not stress-specific
4	AMIGOS	2018	Multimodal emotion dataset	Diverse features & Complex preprocessing
5	Proposed Use	2024	SSL-based physiological dataset	Uses unlabeled data & Requires pretraining

\end{table*}

Table~\ref{tab:dataset} compares several publicly available stress and affect datasets. Each has different strengths: DEAP \cite{b34} and AMIGOS \cite{b35} offer rich EEG-based emotional signals but are not explicitly stress-focused; SWELL-KW \cite{b33} provides naturalistic workplace recordings but limited physiological diversity. We chose WESAD \cite{b10} specifically because it includes the three wrist-worn modalities central to our pipeline (EDA, BVP, TEMP) alongside standardized stress induction, making it the closest available approximation to a real deployment scenario.

subsection{Dataset Description}

WESAD \cite{b10} contains recordings from 15 subjects undergoing a structured lab protocol that alternates between baseline, amusement (watching funny clips), and stress (Trier Social Stress Test) conditions. The chest-worn RespiBAN and wrist-worn EmpaticaE4 provide synchronized multimodal streams; we use the wrist-worn EDA, BVP, and skin temperature channels for all experiments, since these are the modalities available in consumer wearable devices.

Inter-subject variability in WESAD is substantial. Baseline EDA conductance ranges from well under 1 μS to over 20 μS across participants. BVP peak amplitudes differ by more than a factor of three. These differences are biological, not measurement error — and they are precisely why LOSO evaluation produces much lower numbers than random splits. The 15-subject size is admittedly a limitation; results may not hold uniformly across broader populations.

subsection{Data Preprocessing and Preparation}

All signals are preprocessed as described in Section IV: per-subject normalization, low-pass (BVP) and band-pass (EDA) filtering, and fixed-length windowing. Labels follow the WESAD protocol: windows are assigned baseline (0), amusement (1), or stress (2) labels based on condition timing. Windows from transition periods between conditions are discarded to avoid label ambiguity.

For SSL pretraining, labels are ignored entirely. All windows, regardless of stress label, contribute to the unsupervised representation learning phase — which is one of the key practical advantages: unlabeled recordings from future deployments could in principle extend pretraining without any annotation effort.

subsection(Experimental Setup)

All experiments were implemented in Python using PyTorch, trained on a GPU. We used a batch size of 64, which provides sufficient negative samples for contrastive learning without exhausting GPU memory. The Adam optimizer \cite{b1} was used throughout with a learning rate of 0.001; we found that learning rate warmup over the first 5 epochs stabilized contrastive pretraining and prevented early representation collapse. SSL pretraining used 50 epochs; fine-tuning ran for 30 epochs with early stopping. The NT-Xent temperature parameter τ was set to 0.5, matching the original SimCLR recommendation \cite{b4} — lower values sharpen the contrastive objective but risk numerical instability; 0.5 was a reasonable middle ground for the signal lengths we used. The projection head was a two-layer MLP (256-128 dimensions) removed entirely before fine-tuning \cite{b27}.

subsection(Evaluation Metrics)

Four metrics were tracked: accuracy (fraction of correctly classified windows), precision (of all windows labeled stress, how many actually were), recall (of all actual stress windows, how many did we catch), and F1 (their harmonic mean). F1 matters here because the WESAD class distribution isn't perfectly balanced — stress windows make up roughly 35% of the total, so a model that just predicts "not stressed" most of the time could hit decent accuracy without catching much stress at all \cite{b32}. Tracking precision and recall separately also reveals whether errors cluster on the false-positive or false-negative side, which has different clinical implications.



```
\begin{figure}[h]
\centering
\includegraphics[width=\linewidth]{precision.png}
\caption{Precision comparison across different models.}
\label{fig:precision}
\end{figure}
```

Precision results (Figure~\ref{fig:precision}) show the SSL model at 90.8\%, compared to 85.8\% for LSTM. Put differently: of every 100 windows the SSL model flags as stress, about 91 actually are. For LSTM, that figure is 86. The difference matters in clinical use — false stress alerts aren't neutral, they erode user trust in the system.

```
\begin{figure}[h]
\centering
\includegraphics[width=\linewidth]{recall.png}
\caption{Recall comparison of models.}
\label{fig:recall}
\end{figure}
```

Recall (Figure~\ref{fig:recall}) tells the other half of the story: the SSL model catches 91.2\% of actual stress windows, versus 86.2\% for LSTM. So it's missing fewer real stress events. Combined with the high precision, this means the model isn't succeeding by being overly generous with the stress label — it's genuinely more accurate on both sides of the decision boundary.

\subsection{Performance Comparison}

We compare the proposed SSL model against four baselines spanning traditional ML and deep learning, all trained and evaluated under identical LOSO conditions. Figure~\ref{fig:accuracy} summarizes accuracy across all five systems.

```
\begin{figure}[h]
\centering
\includegraphics[width=\linewidth]{accuracy.png}
\caption{Accuracy comparison of different models, showing superior performance of the proposed SSL-based approach.}
\end{figure}
```

\label{fig:accuracy}

\end{figure}

The SSL model reaches 91.5\% accuracy, which is +4.6 points above the LSTM baseline and +13.0 points above SVM. The SVM gap is not surprising — hand-crafted features simply cannot represent the temporal complexity of 60-second physiological windows \cite{b26}. More interesting is the gap over LSTM (86.9\%): both models see the same labeled data during fine-tuning, but the SSL encoder starts from a representation space shaped by hundreds of unlabeled windows, giving it a much better initialization \cite{b16}. The LSTM trained from scratch has to simultaneously learn both low-level signal features and stress-relevant high-level patterns from a limited labeled set — an optimization challenge that SSL sidesteps entirely.

We also note that the performance ordering — SVM $<$ RF $<$ CNN $<$ LSTM $<$ SSL — roughly mirrors the amount of temporal context each model can exploit, which aligns with what we know about stress physiology: the most informative patterns unfold over seconds, not milliseconds.

Performance Comparison of Different Models				
Model	Accuracy	Precision	Recall	F1-Score
SVM	78.5\%	76.2\%	74.8\%	75.5\%
Random Forest	81.3\%	80.1\%	79.5\%	79.8\%
CNN	84.7\%	83.5\%	84.0\%	83.7\%
LSTM	86.9\%	85.8\%	86.2\%	86.0\%
Proposed SSL Model	91.5\%	90.8\%	91.2\%	91.0\%

The results clearly indicate that the proposed SSL-based model outperforms all baseline methods across all evaluation metrics.

Effect of Self-Supervised Learning

What SSL actually changes in this context is the starting point of the encoder, not its architecture. The same CNN-LSTM, when initialized randomly and trained from scratch on labeled data alone, converges to 86.9% accuracy. With SSL pretraining, it reaches 91.5%. The 4.6-point difference reflects representations that were already shaped to be physiologically coherent before the stress labels were introduced \cite{b16}. During fine-tuning, the supervised signal only needs to draw a decision boundary in an already-structured latent space, rather than building that structure from scratch \cite{b31}.

Another way to see this: the LSTM baseline likely overfits subject-specific patterns because it has nothing to learn from except the labeled windows. The pretrained encoder, having seen all the unlabeled data, develops a more distributed representation of physiological variation — one that generalizes because it was never anchored to any particular subject's baseline during pretraining.

subsection{Outcomes of the Cross-Subject Assessment}

Across 15 LOSO folds, performance was relatively consistent — accuracy ranged from a low of around 88% to a high of 94%, depending on the held-out subject. Folds where the model underperformed tended to involve subjects with atypically low EDA reactivity, where the stress-noise separation in the signal was smaller to begin with. This variance is a realistic reflection of individual physiology and is not specific to our model — all baselines showed greater fold-to-fold variability under LOSO than their authors reported under random splits \cite{b29}.

subsection{Stability and Convergence in Training}

Contrastive pretraining loss decreased steadily across all 50 epochs with no signs of representation collapse (a failure mode where all encodings converge to a single point). The SSL initialization led to noticeably faster convergence during fine-tuning — the classification head reached near-optimal accuracy within the first 10 epochs, compared to 20+ for the LSTM trained from scratch. This matters practically: faster fine-tuning means the system can be adapted to a new deployment context with fewer iterations.

subsection{Computational Efficiency}

SSL pretraining does require more total compute than supervised-only training, primarily because each forward pass processes two augmented views per sample \cite{b16}. That said, pretraining is a

one-time overhead; once the encoder is trained, fine-tuning new classifiers for new tasks or populations is fast. Figure-\ref{fig:time} shows training time across all compared models — the SSL overhead is real but bounded, and the pretrained encoder is reusable across tasks, which amortizes the cost over time.

```
\begin{figure}[h]
\centering
\includegraphics[width=\linewidth]{training_time.png}
\caption{Training time comparison across models.}
\label{fig:time}
\end{figure}
```

The learned representations also transfer: an encoder pre-trained for stress detection on WESAD could plausibly initialize a model for a related task such as fatigue or arousal detection, without restarting from random weights \cite{b28}.

\subsection{Discussion of Results}

The confusion matrix (Figure-\ref{fig:cm}) reveals that most misclassifications occur between "baseline" and "amusement" — which makes physiological sense, since both are low-arousal states with similar sympathetic activation profiles. "Stress" is classified most reliably, consistent with its distinctive EDA response signature. This pattern holds across models; the SSL encoder improves absolute performance everywhere, but the relative difficulty order of the three classes remains unchanged.

```
\begin{figure}[h]
\centering
\includegraphics[width=0.8\linewidth]{confusion_matrix.png}
\caption{Confusion matrix of the proposed model showing classification performance.}
\label{fig:cm}
\end{figure}
```

The confusion matrix in Figure-\ref{fig:cm} shows that the model achieves high true positive and true negative rates, indicating effective classification performance.

\section{Discussion}

`\subsection{Ablation Study}`

To evaluate the contribution of different components of the proposed framework, an ablation study is conducted. The performance of the model is analyzed by removing or modifying key components such as self-supervised pretraining and LSTM layers.

`\begin{table}[h]`

`\centering`

`\caption{Ablation Study Results}`

`\begin{tabular}{|c|c|}`

`\hline`

`Model Variant & Accuracy (\%) \\\`

`\hline`

`Without SSL & 84.2 \\\`

`Without LSTM & 86.5 \\\`

`Full Model (Proposed) & \textbf{91.5} \\\`

`\hline`

`\end{tabular}`

`\end{table}`

The ablation numbers (Table above) isolate the contribution of each design choice. Removing SSL pretraining drops accuracy by 7.3 points — which is, frankly, a larger margin than we expected. It confirms that the labeled WESAD data alone is insufficient to train the CNN-LSTM encoder to its full potential; the unlabeled pretraining phase is doing real representational work, not just providing a warm start. Removing the LSTM (keeping only CNN layers) drops accuracy by 5 points, confirming that temporal context across the full window is genuinely informative, not just local waveform shape.

`\subsection{Comparative Study}`

Across the three paradigms surveyed — traditional ML, supervised deep learning, and contrastive SSL — the performance gains align with each method's capacity to exploit temporal context and unlabeled data. As shown in Tables-\ref{tab:paradigm_compare} and-\ref{tab:metric_compare}, our model leads on every reported metric, but the more notable result is the evaluation condition: it is the only method in this comparison that is tested under full LOSO cross-subject holdout.

\cite{b29}. The reported 91.5\% accuracy is therefore conservative relative to the random-split numbers from competing methods.

Per-Metric Performance of Proposed Model vs. Baselines					
Model	Acc.	Prec.	Recall	F1	Avg.
SVM~\cite{h12}	78.5\%	76.2\%	74.8\%	75.5\%	76.3\%
Random Forest~\cite{h26}	81.3\%	80.1\%	79.5\%	79.8\%	80.2\%
CNN~\cite{h13}	84.7\%	83.5\%	84.0\%	83.7\%	83.5\%
LSTM~\cite{h11}	86.9\%	85.8\%	86.2\%	86.0\%	85.9\%
Proposed SSL	91.5\%	90.8\%	91.2\%	91.0\%	90.6\%

Per-metric results in Table~\ref{tab:metric_compare} confirm that the accuracy gap is not an artifact of threshold-tuning: precision (90.8\%) and recall (91.2\%) are both high, producing an F1 of 91.0\% — meaning the model is neither missing most stress events nor flooding the output with false positives \cite{b32}. The contrastive pretraining phase is likely responsible for this balance: by learning to separate signal variation caused by individual differences from signal variation caused by stress state, the encoder produces embeddings where the stress-versus-nonstress boundary is more cleanly separable \cite{b4,b6,b9}.

Figures~\ref{fig:accuracy},~\ref{fig:f1},~\ref{fig:precision}, and~\ref{fig:recall} visualize these trends across all compared models. The gap over LSTM is the most practically meaningful: both models

use the same labeled data for fine-tuning, so the difference is entirely attributable to what the SSL encoder learned during pretraining from unlabeled windows \cite{b16}.

One genuine limitation worth stating plainly: the augmentation choices matter a lot, and we selected them based on physiological intuition rather than a systematic search \cite{b31}. There may be better augmentation combinations — perhaps ones that explicitly model electrode displacement or population-level EDA scaling — that would improve pretraining quality further. This is an open research question.

Ethical Implications and Responsible AI

Deploying a stress detection system outside a research lab raises ethical questions that deserve more than a footnote. Physiological data is among the most sensitive information a person generates — it can reveal anxiety disorders, medication use, and emotional states that the individual may not have chosen to disclose. Any system that passively collects this data in a workplace or healthcare setting must be built with explicit, informed consent mechanisms, not opt-out defaults.

There is also a fairness dimension that the WESAD dataset does not fully address. Its 15 subjects are predominantly young, healthy adults from a single institution. A model trained on this population may not capture the physiological stress signatures of older adults, people with cardiovascular conditions, or individuals on beta-blockers — groups for whom stress monitoring might be most clinically valuable. Deploying the current model to these populations without validation would be irresponsible.

Finally, model outputs should be framed as signals for attention, not clinical diagnoses. Stress classification at 91.5% accuracy still means roughly 1 in 12 predictions is wrong — a rate that is unacceptable if the output drives clinical decisions without human review. The appropriate use of this kind of model is to flag patterns for a clinician or occupational health professional to investigate, not to act on autonomously.

Limitations

Fifteen subjects is a small sample. The LOSO results are encouraging, but WESAD's homogeneity — controlled lab setting, roughly similar demographics — limits how confidently we can extrapolate to the full diversity of people who might use a stress monitoring device. Validation on a larger, more

demographically varied dataset is a necessary next step before any deployment consideration.

The augmentation strategy was chosen based on physiological reasoning and standard wearable signal processing practices \cite{b31}, but not optimized systematically. A poorly chosen augmentation can break the contrastive objective by teaching the encoder to be invariant to information that is actually stress-relevant. An augmentation that changes overall EDA amplitude, too aggressively, for instance, might erase the very conductance shifts that distinguish stress from baseline.

Our current evaluation also assumes relatively clean signals — the preprocessing removes most artifacts, but extreme cases (e.g., electrode detachment, severe motion during recording) are simply excluded rather than handled gracefully. A production system would need a signal quality estimator capable of flagging unreliable windows at inference time.

Applications

A well-validated version of this pipeline could serve several distinct use cases. In clinical settings, continuous stress monitoring via wrist sensor could detect early signs of burnout or anxiety disorder progression between appointments. In workplace wellness programs, aggregate (anonymized) stress trends could alert occupational health teams to high-risk periods — exam seasons, project deadlines, organizational changes. Consumer wearables could use the model to provide biofeedback recommendations ("your stress levels have been elevated for two hours — consider a break"), provided the output is framed appropriately as an indicator rather than a diagnosis.

The methodological framework itself — SSL pretraining on unlabeled physiological data, fine-tuning for a specific affective state — is not specific to stress. The same pipeline could be adapted for fatigue detection, emotional valence tracking, or cognitive load estimation with relatively minor modifications to the fine-tuning head \cite{b28}.

Advantages of Proposed Method

The key practical advantages of this SSL approach over conventional supervised training are:

- Dramatically reduced dependence on labeled physiological data, which is expensive and slow to acquire.

benchmark for wearable stress detection, SSL-pretrained models will likely become the competitive baseline rather than the challenging newcomer. Future work should focus on testing this across larger and more diverse datasets, developing physiological augmentation strategies more systematically, and exploring whether the pretrained encoder can serve as a foundation for a wider range of affective computing tasks.

```
\begin{figure}[h]
\centering
\includegraphics[width=\linewidth]{roc.png}
\caption{ROC curve showing the performance of the proposed model.}
\label{fig:roc}
\end{figure}
```

The ROC curve in demonstrates the model's strong discriminative ability, with a high area under the curve (AUC).

Future Work

A few directions seem genuinely promising and worth naming specifically:

The most obvious extension is multimodal fusion. Voice quality, facial action units, typing rhythm, and app-switching patterns all carry stress information that EDA/BVP/TEMP can't capture \textendash{} there's a ceiling on what wrist-worn physiological sensors can do alone. The tricky part isn't adding modalities; it's handling the mismatched sample rates and the additional failure modes (poor lighting for video, background noise for audio) that compound in real deployment. Early fusion, late fusion, and attention-based fusion are all candidate architectures, and it's not clear a priori which is most robust to missing modalities.

Transformer encoders are worth trying. LSTMs compress sequential information into a fixed-size state, which forces them to forget distant context. Transformers don't \textendash{} they can attend over the full window. Whether that theoretical advantage translates to meaningfully better stress classification on 60-second physiological windows is an empirical question we didn't answer here.

The augmentation strategy was designed by intuition and validated visually. A more principled approach \textendash{} maybe searching over augmentation hyperparameters jointly with the

contrastive objective \textmdash{} might find configurations that extract substantially better representations. There's also the question of whether predictive SSL objectives (masking, forecasting) work better than contrastive ones for physiological signals, or whether hybrid objectives help.

Last, the obvious data limitation: 15 subjects from one institution is not a deployment-ready evaluation. Testing on SWELL-KW \cite{b33} or DEAP \cite{b34}, ideally with the pretrained WESAD encoder used as initialization, would give a much clearer picture of how much the learned representations actually transfer.

\begin{thebibliography}{00}

\bibitem{b1} D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in \textit{Proc. International Conference on Learning Representations (ICLR)}, San Diego, CA, USA, 2015, pp. 1–15.

\bibitem{b2} A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," in \textit{Proc. Advances in Neural Information Processing Systems (NIPS)}, Lake Tahoe, NV, USA, 2012, pp. 1097–1105.

\bibitem{b3} S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," \textit{Neural Computation}, vol. 9, no. 8, pp. 1735–1780, 1997.

\bibitem{b4} T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A Simple Framework for Contrastive Learning of Visual Representations," in \textit{Proc. ICML}, 2020, pp. 1597–1607.

\bibitem{b5} K. He, H. Fan, Y. Wu, S. Xie, and B. Girshick, "Momentum Contrast for Unsupervised Visual Representation Learning," in \textit{Proc. CVPR}, 2020, pp. 9729–9738.

\bibitem{b6} A. van den Oord, Y. Li, and O. Vinyals, "Representation Learning with Contrastive Predictive Coding," \textit{arXiv preprint arXiv:1807.03748}, 2018.

\bibitem{b7} R. Caruana, "Multitask Learning," \textit{Machine Learning}, vol. 28, no. 1, pp. 41–75, 1997.

\bibitem{b8} Y. Bengio, A. Courville, and P. Vincent, "Representation Learning: A Review and New Perspectives," \textit{IEEE Transactions on Pattern Analysis and Machine Intelligence}, vol. 35, no. 8,

pp. 1798–1828, 2013.

[bibitem{b9}] H. Gioreski, M. Gioreski, M. Lutrek, and M. Gams, "Continuous Stress Detection Using a Wrist Device," \textit{IEEE Journal of Biomedical and Health Informatics}, vol. 21, no. 4, pp. 1025–1033, 2017.

[bibitem{b10}] S. Schmidt, B. Reiss, R. Duerichen, and K. Van Laerhoven, "Introducing WESAD: A Multimodal Dataset for Wearable Stress and Affect Detection," in \textit{Proc. ACM ICMI}, 2018, pp. 400–408.

[bibitem{b11}] P. Schmidt, A. Reiss, R. Duerichen, and K. Van Laerhoven, "Wearable-Based Affect Recognition—A Review," \textit{IEEE Access}, vol. 7, pp. 140698–140715, 2019.

[bibitem{b12}] J. Healey and R. Picard, "Detecting Stress During Real-World Driving Tasks Using Physiological Sensors," \textit{IEEE Transactions on Intelligent Transportation Systems}, vol. 6, no. 2, pp. 156–166, 2005.

[bibitem{b13}] S. R. Saha, A. Polatkan, and S. Sarkar, "Deep Learning-Based Stress Detection Using Physiological Signals," \textit{IEEE Access}, vol. 9, pp. 12345–12358, 2021.

[bibitem{b14}] H. Liu, J. Zhao, and Y. Wang, "Multimodal Stress Detection Using Deep Neural Networks," \textit{IEEE Sensors Journal}, vol. 20, no. 15, pp. 8705–8713, 2020.

[bibitem{b15}] M. Zubair, A. Yoon, and J. Kim, "Human Activity Recognition Using LSTM Networks," \textit{IEEE Access}, vol. 8, pp. 12345–12355, 2020.

[bibitem{b16}] X. Wang, R. Girshick, A. Gupta, and K. He, "A Survey on Self-Supervised Learning," \textit{IEEE Transactions on Pattern Analysis and Machine Intelligence}, vol. 44, no. 11, pp. 1–20, 2022.

[bibitem{b17}] Y. LeCun, Y. Bengio, and G. Hinton, "Deep Learning," \textit{Nature}, vol. 521, no. 7553, pp. 436–444, 2015.

[bibitem{b18}] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient Estimation of Word Representations in Vector Space," \textit{arXiv preprint arXiv:1301.3781}, 2013.

[bibitem{b19} A. Vaswani et al., "Attention Is All You Need," in \textit{Proc. NIPS}, 2017, pp. 5998–6008.

[bibitem{b20} D. Silver et al., "Mastering the Game of Go with Deep Neural Networks and Tree Search," \textit{Nature}, vol. 529, pp. 484–489, 2016.

[bibitem{b21} F. Chollet, \textit{Deep Learning with Python}. Manning Publications, 2017.

[bibitem{b22} J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in \textit{Proc. NAACL}, 2019, pp. 4171–4186.

[bibitem{b23} K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," \textit{arXiv:1409.1556}, 2015.

[bibitem{b24} I. Goodfellow et al., "Generative Adversarial Nets," in \textit{Proc. NIPS}, 2014, pp. 2672–2680.

[bibitem{b25} R. Sutton and A. Barto, \textit{Reinforcement Learning: An Introduction}. MIT Press, 2018.

[bibitem{b26} S. Shalev-Shwartz and S. Ben-David, \textit{Understanding Machine Learning: From Theory to Algorithms}. Cambridge University Press, 2014.

[bibitem{b27} S. J. Pan and Q. Yang, "A Survey on Transfer Learning," \textit{IEEE Transactions on Knowledge and Data Engineering}, vol. 22, no. 10, pp. 1345–1359, 2010.

[bibitem{b28} R. W. Picard, \textit{Affective Computing}. MIT Press, 1997.

[bibitem{b29} M. Arlot and A. Celisse, "A Survey of Cross-Validation Procedures for Model Selection," \textit{Statistics Surveys}, vol. 4, pp. 40–79, 2010.

[bibitem{b30} A. Reiss and D. Stricker, "Introducing a New Benchmarked Dataset for Activity Monitoring," in \textit{Proc. IEEE ISWC}, 2012, pp. 108–109.

[bibitem{b31} T. T. Um et al., "Data Augmentation of Wearable Sensor Data for Parkinson's Disease

Monitoring Using Convolutional Neural Networks," in \textit{Proc. ACM ICMI}, 2017, pp. 216–226.

\bibitem{b32} D. M. W. Powers, "Evaluation: From Precision, Recall and F-Score to ROC, Informedness, Markedness and Correlation," \textit{Journal of Machine Learning Technologies}, vol. 2, no. 1, pp. 37–63, 2011.

\bibitem{b33} K. Koldijk, M. Sappelli, S. Verberne, M. A. Neerinx, and W. Kraaij, "The SWELL Knowledge Work Dataset for Stress and User Modeling Research," in \textit{Proc. ACM ICMI}, Istanbul, Turkey, 2014, pp. 291–298.

\bibitem{b34} S. Koelstra et al., "DEAP: A Database for Emotion Analysis Using Physiological Signals," \textit{IEEE Transactions on Affective Computing}, vol. 3, no. 1, pp. 18–31, 2012.

\bibitem{b35} J. A. Miranda-Correa, M. K. Abadi, N. Sebe, and I. Patras, "AMIGOS: A Dataset for Affect, Personality and Mood Research on Individuals and Groups," \textit{IEEE Transactions on Affective Computing}, vol. 12, no. 2, pp. 479–493, 2021.

\end{thebibliography}

\bibliographystyle{IEEEtran}

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\begin{IEEEbiography}[(\includegraphics[width=1in,height=1.25in, clip,keepaspectratio]{a1.png})]{Amit Kumar Bairwa} serves as a distinguished academician in the School of Computer

Science and Engineering at Manipal University Jaipur. He holds an impressive academic background, boasting qualifications such as B.E., M.Tech., and Ph.D. in Computer Science and Engineering. His expertise lies in Optimization, MANET, Network Security, and Machine Learning. As an accomplished author, Dr. Bairwa has published eight books and is actively involved in research across various areas of his interest. He is a valued member of prestigious organizations, including ISTE, ACM, and IEEE, which reflects his commitment to professional growth and networking. In addition to his academic pursuits, Dr. Bairwa is highly proficient in handling real-time challenges related to network and Linux-based systems. He has earned several international certifications, such as CCNA, RHCE, and CEH, demonstrating his dedication to staying current with the latest industry practices. Dr. Bairwa has also been awarded several national and international awards. Dr. Bairwa has significantly contributed to the academic ecosystem during his tenure. He has taken initiatives to enhance the learning environment within the classroom, developed cutting-edge academic and professional programs, and adapted them to meet the evolving needs of the industry in today's dynamic business landscape. He has been awarded with so many titles in the field of teaching and research. His approach involves a well-balanced combination of theoretical concepts and industry-oriented training modules. Dr. Bairwa has collaborated with fellow academicians and played an integral role in setting and maintaining high curriculum standards. He actively contributes to developing mission statements and establishes performance goals and objectives, ensuring that the educational institution consistently excels in its academic pursuits. He also regularly organizes the International Conference on Cyber Warfare, Security & Space Computing (SpacSec) and the International Conference on Computation of Artificial Intelligence & Machine Learning (ICCAIML). Apart from this, He actively conducts workshops and expert talks and gives judgment to top events at the university level.

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